

End Semester Rubric

Q1]

a) For u to be harmonic, it must satisfy Laplace's equation $u_{xx} + u_{yy} = 0$

- Calculating the double derivative:

$$u_x = 3ax^2 + by^2$$

$$u_{xx} = 6ax$$

$$u_y = 2bxy$$

$$u_{yy} = 2bx$$

$$u_{xx} + u_{yy} = 6ax + 2bx$$

equating to zero

$$6ax + 2bx = 0$$

$$\Rightarrow 2x(3a + b) = 0$$

- key step #1 (1)

for this to hold for all x , we need

$$3a + b = 0$$

$$\Rightarrow b = -3a \quad \text{- key step #2 (1)}$$

to find the harmonic conjugate,

we use the Cauchy-Riemann eq's:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$\text{we know, } u_x = 3ax^2 + by^2$$

$$\text{Substituting } b = -3a$$

$$u_x = 3ax^2 - 3ay^2 = v_y$$

Integrating with respect to y

$$v = \int (3ax^2 - 3ay^2) dy$$

$$= 3ax^2y - ay^3 + h(x)$$

- key step #3 (1)

differentiating w.r.t. x

$$v_x = 6axy + h'(x)$$

equating $-v_x$ to u_y

$$\begin{aligned} -6axy + h'(x) &= 2bxy \\ &= -6axy \end{aligned}$$

$$\therefore h'(x) = 0$$

$$\therefore h(x) = C$$

- key step #4 (1)

$$\text{hence, } v(x, y) = 3ax^2y - ay^3 + C$$

- key step #5

(1)

b) We begin with identifying the poles

$$z^2 - 1 = 0$$

$$\Rightarrow z = \pm 1$$

- key step #1

(1)

Our contour is a circle centered

at 1 with radius $R = \pi/2$

then checking which poles lies inside the contour:

for $z=1$

$$|1-1| = 0 < \pi/2$$

$\therefore z=1$ is inside the contour

for $z = -1$

$$|-1-1| = |-2|$$

$$= 2 > 0$$

↖ - key step #2
①

∴ $z = -1$ is outside the contour

setting up & rewriting the integral

$$I = \oint_C \frac{dz}{z^2 - 1}$$

$$= \oint_C \frac{1}{(z-1)(z+1)} dz$$

$$= \oint_C \frac{1/(z+1)}{(z-1)} dz$$

↖ or - key step #3
①

Let $f(z) = \frac{1}{z+1}$, then

by Cauchy integral formula

$$I = 2\pi \cdot i \cdot f(z) \quad \text{— key step #4} \quad \text{①}$$

$$\text{for } z=1, \quad f(1) = \frac{1}{1+1} = \frac{1}{2}$$

$$\therefore I = 2\pi \cdot i \cdot f(1)$$

$$= 2\pi \cdot i \cdot \frac{1}{2} = \pi i \quad \text{— key step #5} \quad \text{①}$$

$\mathbb{Q}_2]$

a)

$$\text{velocity, } v(t) = r'(t)$$

$$= (-a \sin(t))\hat{i} + (a \cos(t))\hat{j} + b\hat{k}$$

$$\text{now acceleration } a(t) = v'(t)$$

$$= (-a \cos(t))\hat{i} - (a \sin(t))\hat{j}$$

— key step #1
①

and speed, $|v| =$

$$\sqrt{(-a \sin(t))^2 + (a \cos(t))^2 + b^2}$$
$$= \sqrt{a^2 + b^2} \quad \text{— key step \# 2 } \textcircled{1}$$

now our tangential component, $a_T =$

$$a_T = \frac{v \cdot a}{|v|}$$

$$\begin{aligned} v \cdot a &= (-a \sin(t))(-a \cos(t)) \\ &\quad + (a \cos(t))(-a \sin(t)) \\ &\quad + (b)(0) \end{aligned}$$

$$\begin{aligned} &= a^2 \sin(t) \cos(t) - a^2 \sin(t) \cos(t) \\ &= 0 \end{aligned}$$

$$\therefore a_T = 0 \quad \text{— key step \# 3 } \textcircled{1}$$

Normal component, $a_N =$

$$a_N = \sqrt{|a|^2 - a_T^2}$$

$$\begin{aligned} \text{here } |a| &= \sqrt{(-a \cos(t))^2 + (-a \sin(t))^2} \\ &= \sqrt{a^2} = a \end{aligned}$$

$$\therefore a_N = \sqrt{a^2 - 0} = a \quad \text{— key step \# 4 } \textcircled{1}$$

$$\begin{aligned} \text{hence, acceleration } a &= 0 \cdot T + a \cdot N \\ &= a \cdot N \end{aligned}$$

$$\text{— key step \# 5 } \textcircled{1}$$

b) For F to be conservative

$$\text{we need } \nabla \times \vec{F} = 0$$

$$\begin{aligned} \therefore (\nabla \times \vec{F})_x &= \frac{\partial}{\partial y} (xy \cos(z)) - \frac{\partial}{\partial z} (x \sin(z)) \\ &= x \cos(z) - x \cos(z) = 0 \end{aligned}$$

$$(\nabla \times F)_y = \frac{\partial}{\partial x} (y \cdot \sin(z)) - \frac{\partial}{\partial x} (xy \cdot \cos(z))$$

$$= y \cdot \cos(z) - y \cdot \cos(z) = 0$$

$$(\nabla \times F)_z = \frac{\partial}{\partial x} (x \cdot \sin(z)) - \frac{\partial}{\partial y} (y \cdot \sin(z))$$

$$= \sin(z) - \sin(z) = 0$$

$\therefore (\nabla \times F) = 0$ ← Shown with steps (2)
(key #1)
hence the field is conservative.

Now, to find the potential function ϕ

$$\phi_x = y \cdot \sin(z)$$

$$\phi_y = x \cdot \sin(z)$$

$$\phi_z = xy \cdot \cos(z)$$

integrating ϕ_x w.r.t. x

$$\phi = x \cdot y \cdot \sin(z) + g(y, z)$$

— key step #2 (1)

differentiating the first term w.r.t. y

$$\frac{\partial}{\partial y} (x \cdot y \cdot \sin(z)) = x \cdot \sin(z) = \phi_y$$

$$\therefore \frac{\partial}{\partial y} [g(y, z)] = 0$$

$$\Rightarrow g(y, z) = h(z)$$

— key step #3 (1)

differentiating $(x \cdot y \cdot \sin(z))$ w.r.t. z

$$\frac{\partial}{\partial z} (x \cdot y \cdot \sin(z)) = x \cdot y \cdot \cos(z) = \phi_z$$

$$\therefore \frac{d}{dz} [h(z)] = 0$$

$$\Rightarrow h(z) = C$$

∴ potential function $\phi(x, y, z) = x \cdot y \cdot \sin(z) + C$

- key step #4

①

Q3]

a)

$$\text{Let } P = \frac{-y}{x^2 + y^2}$$

$$Q = \frac{x}{x^2 + y^2}$$

$$R = \frac{z}{x^2 + y^2}$$

$$\text{curl } F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$\begin{aligned} \hat{i} \text{ component} &= \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \\ &= 0 - 0 = 0 \end{aligned}$$

as R does not depend on y &
 Q does not depend on z

$$\begin{aligned} \hat{j} \text{ component} &= \frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \\ &= 0 - 0 = 0 \end{aligned}$$

as R does not depend on x &
 P does not depend on z

Now we are only left with the $\hat{k}$ component.

Showing
 $\hat{i}$ & $\hat{j}$
component
as zero
using any
method
(key step
#1)

①

$$(\text{Curl } F)_k = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

calculating
with steps
(key step
2)
②

$$\begin{aligned} \frac{\partial Q}{\partial x} &= \frac{1(x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2} \\ \frac{\partial P}{\partial y} &= \frac{-1(x^2+y^2) - (-y)(2y)}{(x^2+y^2)^2} \\ &= \frac{-x^2-y^2+2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2} \end{aligned}$$

subtracting gives 0

$$\therefore \text{Curl } F = 0$$

- evaluating the final curl = 0
with any formula used

(key step #3) ①

b) We begin with parameterizing the curve, as we have a unit circle, we use the polar coordinates

$$\therefore \text{ say } x = \cos(t)$$

$$y = \sin(t)$$

$$z = 0$$

$$\text{where } t \in [0, 2\pi]$$

Now

$$dx = -\sin(t) dt$$

$$dy = \cos(t) dt$$

$$dz = 0$$

key step # 1

①

Now

$$I = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

$$= \int_0^{2\pi} [P dx + Q dy + R dz]$$

- key step # 2 : decomposing $\mathbf{F} \cdot d\mathbf{r}$
to get $\downarrow$ (could be done before)
integral (2)

$$= \int_0^{2\pi} [(-\sin(t))(\sin(t) dt) + (\cos(t))(\cos(t) dt) + 0]$$

$$= \int_0^{2\pi} [\sin^2(t) + \cos^2(t)] dt$$

$$= \int_0^{2\pi} 1 dt \quad \text{--- key step # 3} \quad (1)$$

c) No. It Stokes's Theorem is not contradicted.

--- key statement # 1 (1)

$$\text{Stokes's Theorem} = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$$

However for this theorem to hold, the partial derivatives must be continuous everywhere on the surface bounded by the curve C

If our surface is the unit disk ($x^2 + y^2 \leq 1$) that includes the point $(0, 0)$

Here the denominator of $\mathbf{F}$ goes to zero $\therefore$ the field is undefined at origin $(0, 0)$

--- key statement # 2 (1)

$\therefore$ the domain is not simply connected hence Stokes's theorem just cannot be applied.

Q4]

- a) The surface in question is part of the plane defined by the equation $x+y+z=1$

we can rewrite this explicitly as x & y

$$z = f(x, y) = 1 - x - y$$

The region of integration D is determined by the intersection with the cylinder $x^2 + y^2 = 1$.

$\therefore$ our domain D is the unit disk in the xy plane:

$$D = \{ (x, y) \mid x^2 + y^2 \leq 1 \}$$

The area S of a surface $z = f(x, y)$ over a region D is given by the double integral:

$$S = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

• differentiating $z = 1 - x - y$ w.r.t. x & y :

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (1 - x - y) = -1$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} (1 - x - y) = -1$$

} key step #1
①

substitute back to the integral:

$$S = \iint_D \sqrt{1 + (-1)^2 + (-1)^2} \, dA$$

$$= \iint_D \sqrt{3} \, dA$$

— key step # 2

(with the 2 integral formula from before)

$$= \sqrt{3} \iint_D dA$$

$\iint_D dA$ is the area of the domain D

which is a unit disk with radius, $r=1$
it's area is known from geometry

$$\text{Area}(D) = \pi r^2 = \pi(1)^2 = \pi$$

— key step # 3 ①

$$\therefore S = \sqrt{3} \cdot \pi$$

$$\text{hence, area} = \pi\sqrt{3}$$

— key step # 4 ①

b) * Using the parametric surface method:

Since the surface is a paraboloid

$$z = x^2 + y^2$$

we use the polar coordinates as our parameters $u = r$ & $v = \theta$

$$\begin{aligned}\text{let } x &= r \cos(\theta) \\ y &= r \sin(\theta) \\ z &= r^2\end{aligned}$$

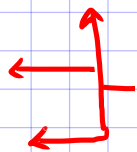
The position vector is

$$r(r, \theta) = \langle r \cos(\theta), r \sin(\theta), r^2 \rangle$$

with

$$0 \leq r \leq 1$$

$$\& \quad 0 \leq \theta \leq 2\pi$$



key step # 1

①

The tangential vectors are given by
the partial derivatives of r w.r.t. r & θ

$$r_r = \langle \cos(\theta), \sin(\theta), 2r \rangle$$

$$r_\theta = \langle -r \sin(\theta), r \cos(\theta), 0 \rangle$$

— key step # 2

①

The normal vector

$$N = r_r \times r_\theta$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos(\theta) & \sin(\theta) & 2r \\ -r \sin(\theta) & r \cos(\theta) & 0 \end{vmatrix}$$

$$= \langle -2r^2 \cos(\theta), -2r^2 \sin(\theta), r \rangle$$

the $\hat{k}$ component is r (+ve) which points
up/inwards.

We must reverse the $\hat{k}$ component for
flux outwards (away from z)

$$\therefore N_{\text{out}} = -N = \langle 2r^2 \cos(\theta), 2r^2 \sin(\theta), -r \rangle$$

- key step # 3 ①

$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{N}_{\text{out}} \, dA$$

here $\mathbf{F} \cdot \mathbf{N}_{\text{out}} = \langle 4x, 4y, 2 \rangle \cdot$

$$\langle 2r^2 \cos(\theta), 2r^2 \sin(\theta), -r \rangle$$

$$= \langle 4r \cos(\theta), 4r \sin(\theta), 2 \rangle \cdot$$

$$\langle 2r^2 \cos(\theta), 2r^2 \sin(\theta), -r \rangle$$

$$= (4r \cos(\theta))(2r^2 \cos(\theta))$$

$$+ (4r \sin(\theta))(2r^2 \sin(\theta))$$

$$+ (2)(-r)$$

$$= 8r^3 \cos^2(\theta) + 8r^3 \sin^2(\theta) - 2r$$

$$= 8r^3 (\cos^2(\theta) + \sin^2(\theta)) - 2r$$

$$= 8r^3 - 2r$$

- key step # 4 ①

substituting in the flux integral

$$\text{Flux} = \int_0^{2\pi} \int_0^1 (8r^3 - 2r) \, dr \, d\theta$$

$$= 2\pi \left[\frac{8r^4}{4} - \frac{2r^2}{2} \right]_0^1$$

$$= 2\pi [2(1)^4 - (1)^2]$$

$$= 2\pi (1) = 2\pi$$

- key step # 5 ①